

Lab 07

BJT as Switch and Emitter-Follower Amplifier

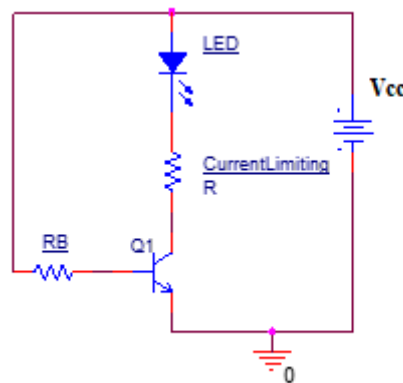
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7.1 Objective

To investigate the operation of BJT common-collector (emitter-follower) amplifier and design a circuit for LED control using BJT as a switch.

Note: Pre-lab (printed or hand-written) should be done on a separate paper (use **A4 sheet**) and will be submitted at the start of Lab within first 10 minutes.

7.2 Pre-Lab Task



Task 1: To derive expressions for designing BJT switch controlling LED.

- For the circuit given above derive the expressions for the base resistor R_B and the current limiting resistor R using KCL.
- Given that the forward voltage V_{LED} and current through LED is 2V and 20mA respectively, what will be the value of collector current I_C .

Lab 07

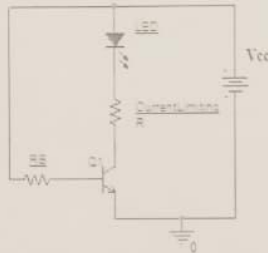
BJT as Switch and Emitter-Follower Amplifier

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7.1 Objective

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$$\begin{aligned}
 I_C &= I_{LED} \\
 I_C &= 20\text{mA} \text{ (Since IC is connected with the LED in the collector branch in series)} \\
 V_{RC} &= I_C \times R_C \\
 V_{CC} &= V_{RC} + V_{LED} + V_{CE} \\
 V_{CC} &= I_C \times R_C + V_{LED} + V_{CE} \\
 R_C &= (V_{CC} - V_{LED} - V_{CE}) / I_C \\
 V_{in} &= V_{RB} + V_{BE} \\
 V_{RB} &= I_B \times R_B \\
 I_C &= \beta \times I_B \\
 I_B &= I_C / \beta
 \end{aligned}$$

7.3 Introduction to Emitter-Follower Common Collector Amplifier

When the output of BJT amplifier is taken from the emitter terminal of the transistor as shown in Figure 1, the network is referred to as an emitter-follower. This configuration is actually a common-collector configuration because the collector is directly connected to supply and this will be grounded for AC analysis. The emitter-follower configuration is frequently used for impedance-matching purposes.

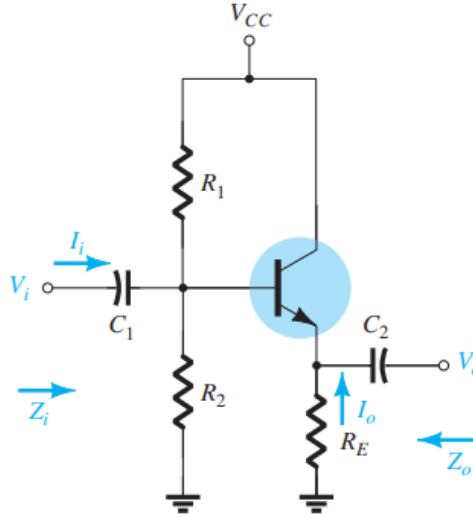


Figure 1: Emitter-follower amplifier circuit

For the circuit shown in Figure 1, if BJT is replaced by its r_e model, the AC equivalent of the network can be represented by Figure 2. Here, R' is the equivalent impedance $R1 \parallel R2$.

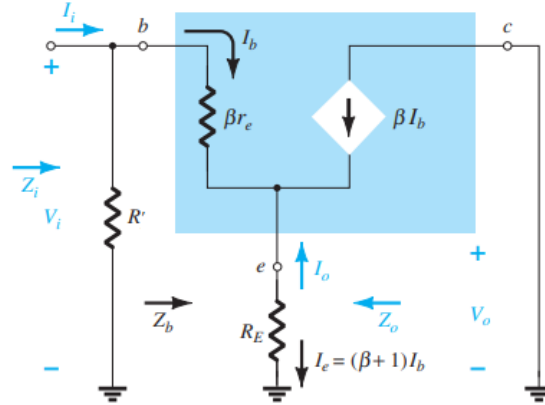


Figure 2: AC equivalent of the emitter-follower network

7.4 Characteristic Parameters of Emitter-Follower CC Amplifier

Using the network of Figure 2, the characteristic parameters of emitter-follower circuit are given as:

- AC Voltage Gain A_v is defined as $A_v = \frac{V_o}{V_i}$

Here, V_o and V_i are output and input voltage which can be measured as rms, peak or peak-to-peak. From model, the AC voltage gain of a CC amplifier (under no-load) can be calculated using:

$$A_v = \frac{R_E}{R_E + r_e}$$

Where, transistor AC dynamic resistance r_e shown in model can be calculated as

$$r_e = \frac{V_T}{I_E} = \frac{26mV}{I_E(mA)}$$

- AC Input Impedance, Z_i , is that of the amplifier as seen by the input signal. For CC amplifier, it can be given as

$$Z_i = R_1 \parallel R_2 \parallel \beta(R_E + r_e)$$

- AC Output Impedance, Z_o , for CE amplifier can be approximated to

$$Z_o = r_e$$

Task 2: To determine DC operating point of transistor for emitter-follower amplifier

- Implement the circuit shown in Figure 3 using 2N3904 BJT model and supply of 10V.
- Note down the measured values of resistances used in your implemented circuit.
- Measure the DC operating values of the circuit as given in Table 1.

Table 1: DC operating point of BJT in CC amplifier circuit

Parameters	Measured Value	Parameter	Measured Value	Parameter	Measured Value
V_{CC}	10V	Base Voltage: V_B	2.23V	Base Current: I_B	7.6u
R_1	33.04k	Emitter Voltage: V_E	2.53V	Collector Current: I_C	1.599m
R_2	11.5k	Collector Voltage: V_C	10.035V	Emitter Current: I_E	1.66m
R_E	0.98k	Base-Emitter Voltage: V_{BE}	0.66V	DC Current Gain: β	210.395

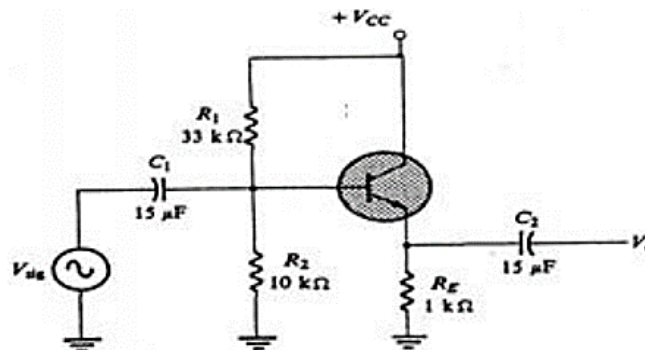


Figure 3: Emitter-Follower CC amplifier circuit for implementation

Task 3: To calculate the characteristic parameters of emitter-follower amplifier

- Now, using the measured values of Table 1, calculate the characteristic parameters of amplifier listed in Table 2.

Table 2: Characteristic parameters of CC amplifier circuit

Parameters	Calculated Values
Transistor AC dynamic resistance r_e	15.66
AC Voltage Gain A_v	0.9843

Input Impedance Z_i	8.1969k
Output Impedance Z_o	15.66

Task 4: To experimentally determine characteristic parameters of emitter-follower CC amplifier

1. Complete the amplifier implementation, and apply AC input signal V_{sig} of 1Vrms (1.414V peak) at frequency of 1k Hz at no load condition through coupling capacitor C1. Measure the output voltage through C2 at no-load. Make sure there is no distortion or clipping of waveform at the output terminal emitter.

• Amplifier Voltage Gain: A_v

1. Obtain the waveforms of input and output voltage. Note down their measured values in Table 3. Determine the gain of amplifier.
2. Observe the phase difference between the two and assign corresponding sign (+/-) to the gain.

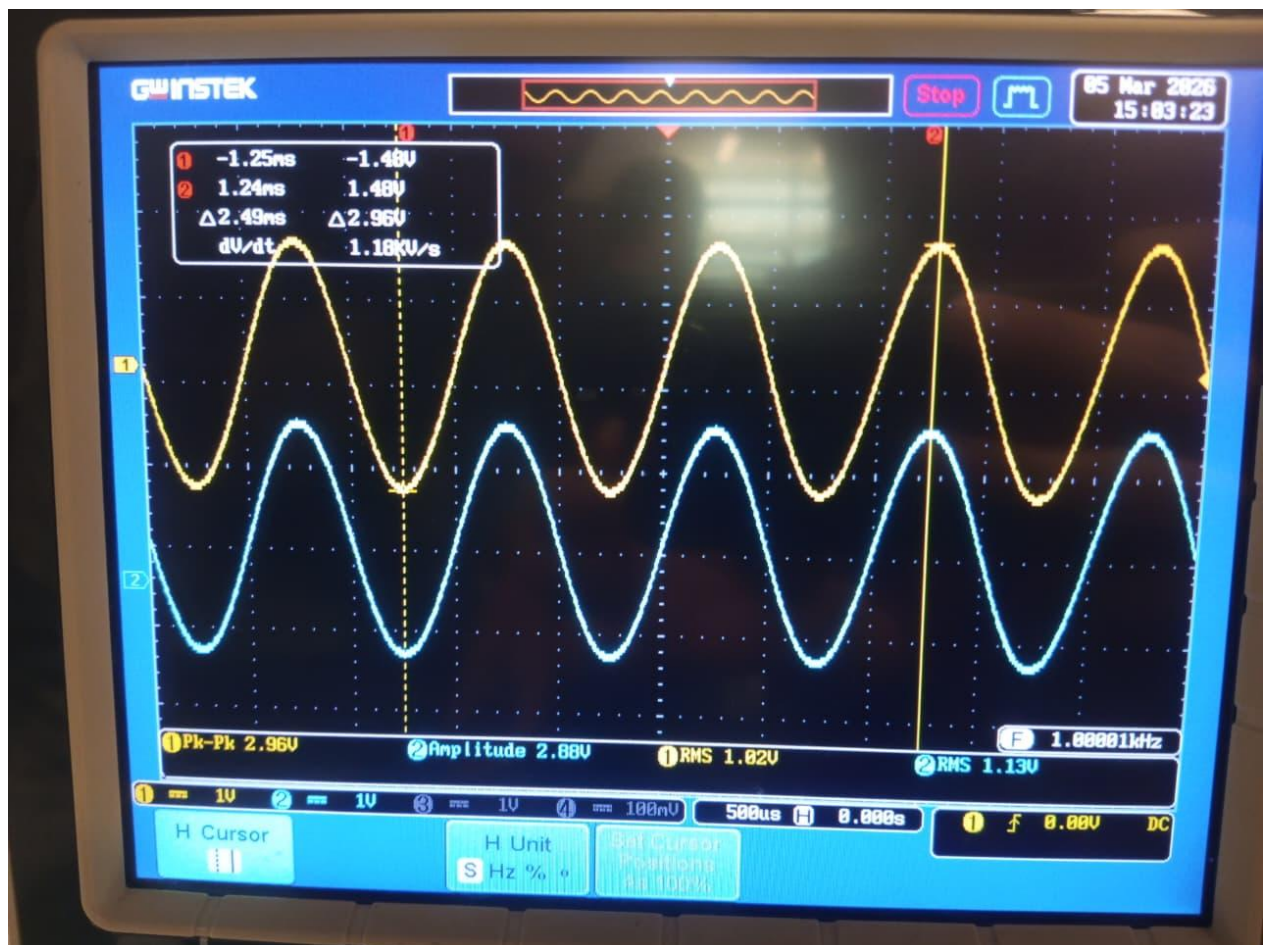
When the input and output waveforms were observed on the oscilloscope, both signals were in phase with each other. The peaks and zero crossings of the input signal occurred at the same time as the output signal. Because there is no phase inversion, the voltage gain is assigned a positive sign (+). This behavior is expected for a common-collector (CC) amplifier, since it does not invert the signal like the common-emitter amplifier.

3. What are your comments about the gain of amplifier? Keeping the formula of gain in terms of circuit parameters, do you think it is easily adjustable? Justify the name of this CC amplifier as *emitter-follower*.

From the measured values, the voltage gain of the amplifier was found to be approximately 1. which means the output voltage is almost equal to the input voltage but a little smaller due to the base emitter voltage drop and internal transistor resistance.

According to the gain expression of the CC amplifier, the gain mainly depends on R_E and the small signal emitter resistance r_e . Since r_e is very small compared to R_E , the gain naturally becomes very close to unity. Because of this relationship, the gain is not easily adjustable and generally remains close to 1 for most practical values of the circuit components.

This is why the CC amplifier is called an emitter follower. The output voltage taken from the emitter follows the input voltage applied at the base, with almost the same waveform and amplitude. The emitter voltage simply tracks the base voltage, which justifies the name emitter follower.



• Amplifier Input Impedance: Z_i

1. To determine the input impedance of this amplifier, connect an input measurement resistor, $R_x=10\text{k}\Omega$, as shown in Figure 4. Apply the AC input signal of 2V_{rms} at frequency of 1 kHz but this time through this known resistance. Measure V_i which is the signal after this resistor R_x input to amplifier. Note down in Table 3.

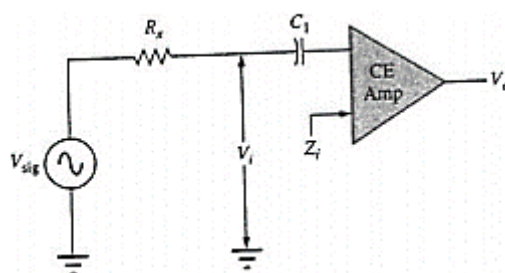


Figure 4: Circuit for determining input impedance

1. Mathematically, for this circuit this voltage V_i can now be expressed as

$$V_i = \frac{V_{sig}}{Z_i + R_x} Z_i$$

Solve the given equation for Z_i .

$$Z_i = \frac{V_i R_x}{V_{sig} - V_i}$$

- Determine the value of input impedance. Note down in Table 3.

Table 3: Experimentally determined values of CC amplifier characteristic parameters

Characteristic Parameter	Experimentally Determined Values	Required Measurements	Measured Values
CC Amplifier Voltage Gain: A_v	0.989	Input Voltage: V_i	760 mVrms
		Output Voltage (Unloaded): V_o	752 mVrms
CC Amplifier Input Impedance: Z_i	7.161 kOhms	Measurement Resistor: R_x	9.8k
		Source Voltage: V_{Sig}	1.8Vrms
		Input Voltage: V_i	760 mVrms
CC Amplifier Output Impedance: Z_o	21.94 Ohms	Output Voltage Unloaded: V_o	104 mVrms
		Load Resistor: R_L	98.18
		Output Voltage Load: V_L	80.5 mVrms

• Amplifier Output Impedance: Z_o

- Remove the input measurement resistor R_x .
- Apply AC input signal (100 mV rms at 1 kHz) directly. Measure the unloaded output voltage V_o .
- Now, connect load resistor of 100 Ω and measure the corresponding loaded output voltage V_L . Note it down in Table 3.
- When loaded, the output voltage can be expressed as

$$V_L = \frac{R_L}{Z_o + R_L} V_o$$

Solve the above for output impedance Z_o .

$$Z_o = \frac{R_L(V_o - V_L)}{V_L}$$

- Calculate the output impedance to complete Table 3.

7.5 Transistor as a Switch

Transistor can be used as a switch to control the ‘on’ and ‘off’ states of a load. The behavior of switch is represented in Figure 5. When a switch is off, it can be replaced by open circuit and when it is fully on, it can be replaced by short-circuit.

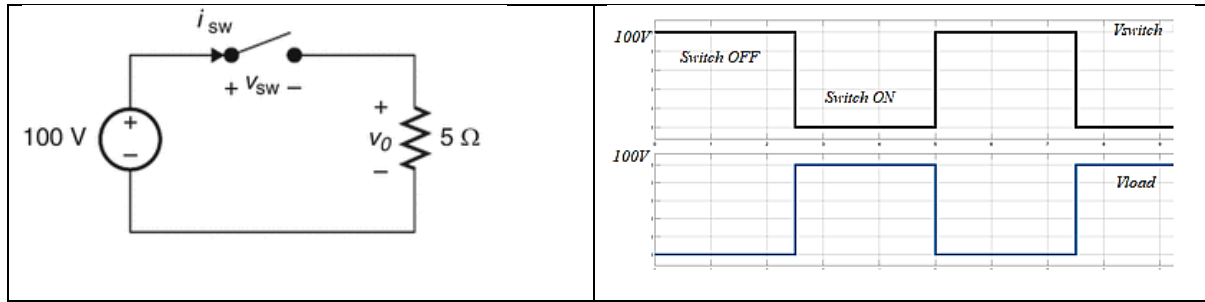


Figure 5: Characteristics of ideal switch

For implementation of switch by transistor, the transistor operates in saturation region and cut-off region. In cut-off region, both the junctions remain reversed-biased so it acts as “fully-OFF” and in saturation region, both the junction remain forward-biased so it acts as “fully-ON”. In common-emitter configuration, the voltage across collector-emitter terminal is equal to the supply voltage V_{CC} in cut-off region and in saturation region, collector-emitter voltage remains approximately equal to zero (0.3 V). Therefore, the base terminal can be used to control the operation of transistor switch that can control the state of load connected at collector terminal. **The collector and emitter form the switch terminals and the base is the switch handle.** In other words, the small base current can be made to control a much larger current between the collector and emitter. For example, the circuit of Figure 6 can be modified to control an electric motor. The BJT switch only needs to handle the small base current due to the current amplification provided by the transistor. If high current motors are directly switched on and off, mechanical switch contacts will eventually wear out from switching the high current, causing the switch to fail. However, the BJT can operate nearly indefinitely as the switch as it has no mechanism that causes it wear out. The loads that require a few milliamps can be directly controlled by connecting them at collector terminal. For devices that require more power like large motors, the transistor switch can be used to control relay (as relay driver) that controls motor.

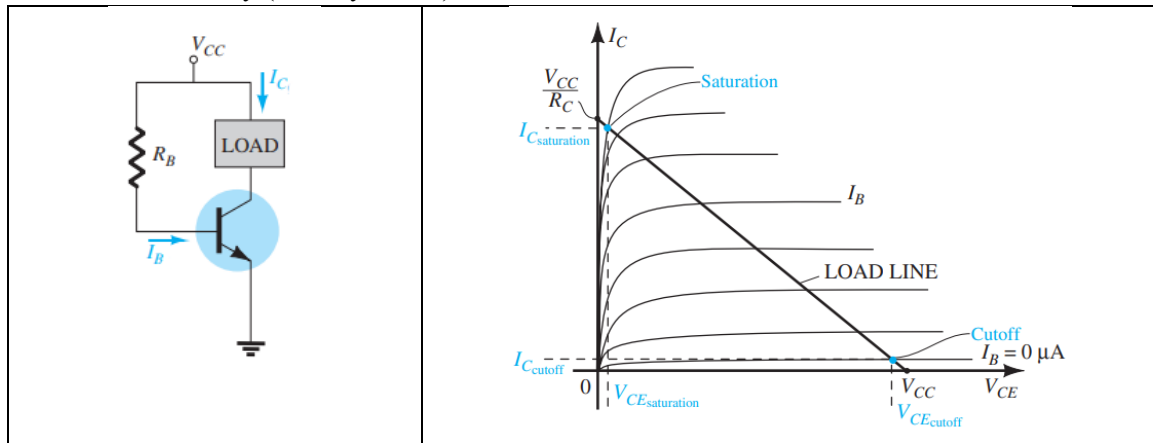


Figure 6: Transistor as switch

When EBJ is forward biased, let V_{BE} be 0.7 V. We know that when collector voltage is lower than the base voltage by 0.4V, the transistor enters saturation region. For that edge of saturation (EOS) point, we can take V_C equal to 0.3V. For a transistor that operates deeper into saturation region, we take V_{CE_sat} equal to 0.2V.

$$I_{C_Sat} = \frac{V_{CC} - V_{CE_Sat}}{R_C}$$

Using minimum value of current gain β_{min} , corresponding edge-of-saturation base current I_{B_EOS} required to saturate the transistor is given by

$$I_{B_Sat} = \frac{I_{C_Sat}}{\beta_{min}}$$

In saturation region, the transistor can be forced to operate at any desired β . The ratio of collector (in saturation) to base current is therefore, called β_{forced} . Ratio of I_B to I_{B_Sat} is called overdrive factor. This can be used to calculate required base current and resistance R_B .

$$V_{BB} - V_{BE} = I_B R_B$$

Task 5: To design circuit for implementing BJT switch to control an LED

To design a circuit that utilizes transistor as a switch for controlling *on* and *off* status of Red LED, consider the circuit shown in Figure 7. Here, in series with LED, there is a current limiting resistor R . The LED forward voltage is 2V and has forward current rating of 20mA. The LED with current limiting resistor will act as load across which 5V should appear when the switch is on and 0V when switch is *off*.

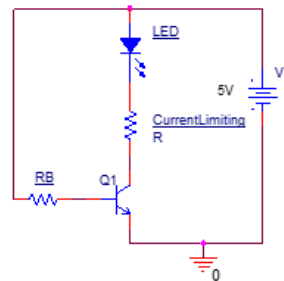


Figure 7: Implementation of transistor switch to control LED Find the value of current limiting resistor R and base resistor R_B with β_{min} of 100 and an overdrive factor of at least 10. Show your calculations. *Hint: Use equations derived in Task 1.*

**Given Information:**

- Supply voltage: $V_{CC} = 5V$
- LED forward voltage: $V_{LED} = 2V$
- LED forward current rating: $I_{LED} = 20mA$
- $\beta_{min} = 100$
- Overdrive factor = 10
- $V_{BE} = 0.7V$ (when forward biased)
- $V_{CE_{sat}} = 0.2V$ (transistor in saturation)

Step 1: Calculate Current Limiting Resistor R

When the transistor switch is ON (saturated), the LED and resistor R are in series between V_{CC} and the collector. Using Kirchhoff's Voltage Law (KVL) around the collector loop:

$$V_{CC} = V_{LED} + V_{resistor} + V_{saturation}$$

Where V_R is the voltage across resistor R.

Solving for V_R :

- $V_R = V_{CC} - V_{LED} - V_{CE_{sat}}$
- $V_R = 5V - 2V - 0.2V$
- $V_R = 2.8V$

The collector current equals the LED current (since they're in series):

- $I_C = I_{LED} = 20mA = 0.02A$

Using Ohm's Law to find R:

- $R = V_R / I_C$
- $R = 2.8V / 0.02A$
- $R = 140\Omega$

Step 2: Calculate Base Current at Edge of Saturation (EOS)

The minimum base current needed to just saturate the transistor is:

$$I_{B(EOS)} = I_C / \beta_{min}$$

- $I_{B(EOS)} = 20mA / 100$
- $I_{B(EOS)} = 0.2mA = 200\mu A$

Step 3: Apply Overdrive Factor to Calculate Actual Base Current

To ensure the transistor operates deep in saturation (not just at the edge), we apply an overdrive factor:

$$I_B = \text{Overdrive Factor} \times I_{B(EOS)}$$

- $I_B = 10 \times 0.2mA$
- $I_B = 2mA = 0.002A$

This ensures $\beta_{forced} = I_C / I_B = 20mA / 2mA = 10$, which is much lower than $\beta_{min} = 100$, guaranteeing saturation.

Step 4: Calculate Base Resistor R_B

When 5V is applied at the input (switch ON), using KVL in the base loop:

$$V_{in} = I_B \times R_B + V_{BE} \text{ (from prelab)}$$

Solving for R_B :

- $R_B = (V_{in} - V_{BE}) / I_B$
- $R_B = (5V - 0.7V) / 0.002A$
- $R_B = 4.3V / 0.002A$
- $R_B = 2.15k\Omega$

We can use a standard resistor value of $2.2k\Omega$ (closest standard value).

Task 6: To implement and investigate operation of BJT switch for LED control

- Construct the circuit designed in Task 5 using 2N3904 BJT.
- Identify the modes of operation of transistor when voltage is applied at base terminal through R_B and when the applied voltage at this terminal is 0V. Don't change the voltage V_{CC} applied at load. Observe the status of LED Measure the voltage across the load (LED + Resistor) and transistor switch in each case along with base current and collector current.

Mode of Operation	(Saturation region) @ $V_{CC}=5V$	(Cut off region) @ $V_{CC}=0V$
$V_{LOAD} = V_{(LED+R)}$	4.934V	-18.67V
I_C	17.42mA	0uA
I_B	1.90mA	0.20uA

- Now, instead of controlling transistor switch through constant supply at base terminal, control it through a pulse signal with 50 % duty cycle having amplitude of 5V. Use the function generator to generate 1Hz square wave of 0-5V (5V peak to peak and offset of 2.5V). Why is offset added?

Instead of applying constant DC voltage to the base, the transistor was controlled using a square wave pulse signal from the function generator.

The signal used was:

- Frequency: 1 Hz
- Amplitude: 5 V peak-to-peak
- Offset: 2.5 V

The offset is added so that the signal varies between 0V and 5V. Without the offset, the square wave would swing between +2.5V and -2.5V, which would apply negative voltage to the base and could damage the transistor or cause incorrect operation.

With the offset:

- LOW level = 0V → transistor OFF
- HIGH level = 5V transistor ON

Because of this, the LED turns ON and OFF periodically, following the input pulse.

- Initially set frequency of 1Hz and then increase it. Does the transistor switch keep changing its status at higher frequencies? You can verify this by observing glow of LED and obtaining voltage waveform across transistor switch / load.

Initially at 1 Hz, the LED clearly turned ON and OFF every second.

As the frequency was increased:

- The transistor continued switching correctly.
- However, at higher frequencies the LED appeared continuously ON.

This happens because the human eye cannot detect very fast blinking, so the LED looks constantly lit even though it is still switching.



7.6 Post Lab Task

Task 7: To analyze the performance of BJT as amplifier and switch

1. Discuss the application of emitter-follower circuit as voltage buffer.

Application of Emitter-Follower as a Voltage Buffer

The emitter-follower circuit is commonly used as a voltage buffer. In this configuration, the output voltage at the emitter is almost equal to the input voltage applied at the base (slightly lower by about 0.7V).

The main advantage of this circuit is that it provides:

- Very high input impedance
- Very low output impedance

Because of this property, it can connect a high impedance source to a low impedance load without significantly affecting the signal. In other words, it prevents loading effects and allows efficient signal transfer between stages.

For example, it is often used between sensor circuits and signal processing circuits, or between amplifier stages.

2. Compare characteristic parameters of CE amplifier and CC amplifier.

Parameter	CE Amplifier	CC Amplifier (Emitter Follower)
Voltage Gain	High (>1)	Approximately 1
Phase Shift	180° (inverted)	0° (no inversion)
Input Impedance	Moderate	Very High
Output Impedance	High	Very Low
Main Application	Voltage amplification	Voltage buffering / impedance matching



Assessment Rubric
Lab 07
BJT as Switch and Emitter-Follower Amplifier

Name:	Student ID:
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Points Distribution

Task No.	LR1 Circuit Layout	LR 4 Data Collection	LR 6 Calculation	LR 10 Analysis	LR 11 Design
Task 1	-	-	-	-	/8
Task 2	/4	/10	-	-	-
Task 3	-	-	/8	-	-
Task 4	/4	/12	/4	/8	-
Task 5	-	-	-	-	/8
Task 6	/4	/8	-	/8	-
Task 7	-	-	-	/10	-
	/12	/30	/12	/26	/16
CLO Mapped	CLO 1	CLO 1	CLO 1	CLO 1	CLO 2

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/4	CLO 4

CLO	Total Points	Points Obtained
1	80	
2	16	
4	04	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.